

SupportPearlz Technical Report

LangChain Real-World RAG Customer Support Agent

Prototype implementation based on the supplied AAI-407 / PRAC-03 assignment manual.

1. Architecture and design

The implementation separates ingestion from query time. Ingestion walks the knowledge-base folder, dispatches loaders by extension, normalises text, attaches source metadata, chunks documents, embeds chunks and persists them in Chroma. The CLI only loads the persisted index and performs retrieval and generation. This avoids re-embedding the corpus for every question.

2. Dataset

The repository contains 13 fictional knowledge-base files using Markdown, TXT and CSV. The loader layer also supports PDF and DOCX as required. The dataset is internally consistent around a 24-month warranty, a 14-day return window, remote-area delivery to Skardu in 5–7 working days, and documented filter pricing. Three cross-document seams and three deliberate gaps are documented in `gaps_and_seams.md`.

3. Chunking experiment

Six configurations are generated by the experiment helper: 300/0, 300/60, 800/0, 800/120, 1500/0 and 1500/225 characters/overlap. The final default is 800/120 because it balances policy-clause precision with enough local context for grounded answers. The exact measured values should be regenerated after the user's environment builds the index; the repository deliberately does not fabricate retrieval measurements.

4. Embeddings and vector store

The default configuration uses OpenAI text-embedding-3-small for both indexing and querying, with Chroma persisted on disk. `embedding_diagnostic.py` embeds two paraphrases and an unrelated sentence and prints pairwise cosine similarity. The same embedding model is required for both sides of the search space. Changing it requires a full rebuild.

5. Retrieval

Similarity search is the default. `k`, threshold and MMR fetch size are environment-driven. A relevance gate removes hits below the configured threshold; the threshold must be calibrated on the user's own score distribution rather than treated as a universal number. MMR is available for diversity when multi-document questions benefit from less redundancy.

6. Grounding and structured output

The final prompt explicitly requires evidence from the supplied context, refuses unsupported claims, handles partial answers, validates citations and treats instructions inside retrieved text as data. The response is validated with Pydantic. Invalid or unavailable generation falls back to a user-readable failure response instead of exposing a traceback.

7. Conversational memory

ConversationMemory keeps a bounded number of turns and uses a dedicated condensation call to rewrite a follow-up such as “Does that apply if I opened it?” into a standalone question. The original and rewritten queries are logged. A topic switch is allowed to replace stale context because the latest user question is passed to the condensation model with recent history.

8. Evaluation

evaluation/test_questions.json contains 20 cases spanning easy, multi-document, follow-up, unanswerable, ambiguous, adversarial and robustness categories. run_eval.py records raw answers and three lightweight scores. Manual review remains necessary for true groundedness and hallucination count; the project intentionally does not claim a perfect score without evidence.

9. Limitations and next steps

The prototype has no live order, courier, payment or ticketing tool. MMR scores are displayed neutrally because the selected API does not expose the same relevance scores. The evaluation heuristic is intentionally conservative. A production version should add FastAPI, authentication, source-version lifecycle management, PII redaction, regression gates, tracing, caching and an independent reranker.

Reproducibility note

The supplied assignment requires actual measured chunking/retrieval results and genuine screenshots. This package includes capture templates rather than pretending that live API execution occurred in this environment. Run the commands in README.md, then replace the templates with real evidence before submission.